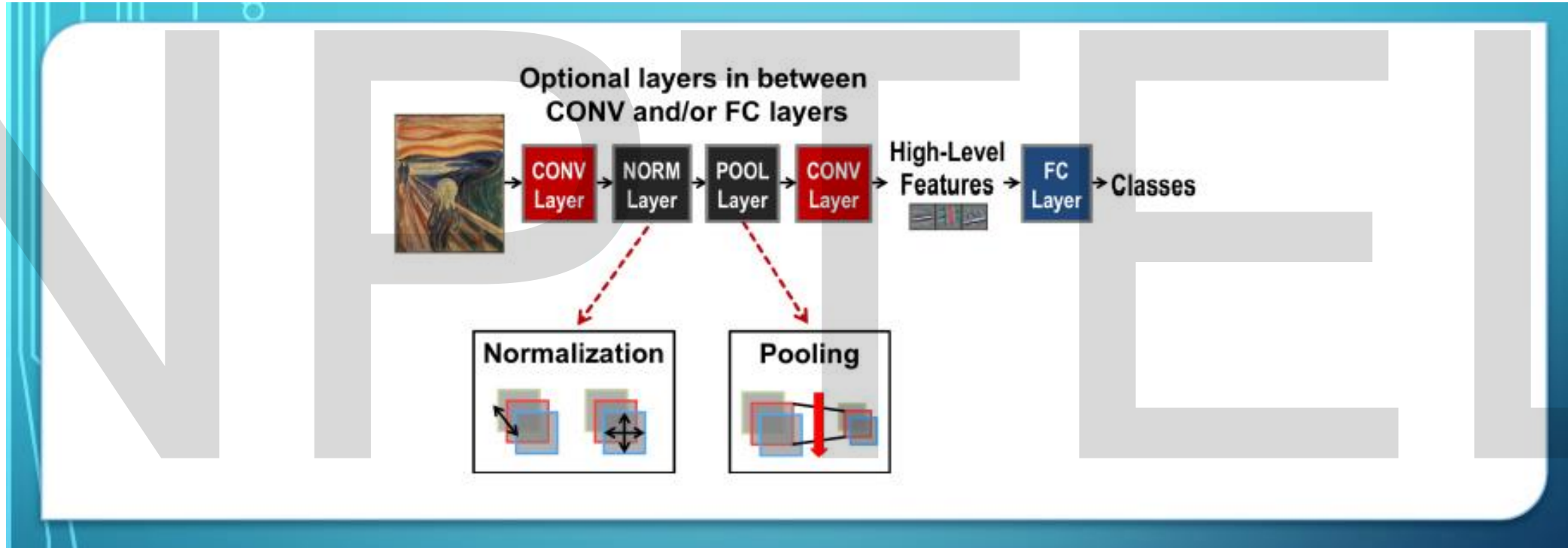
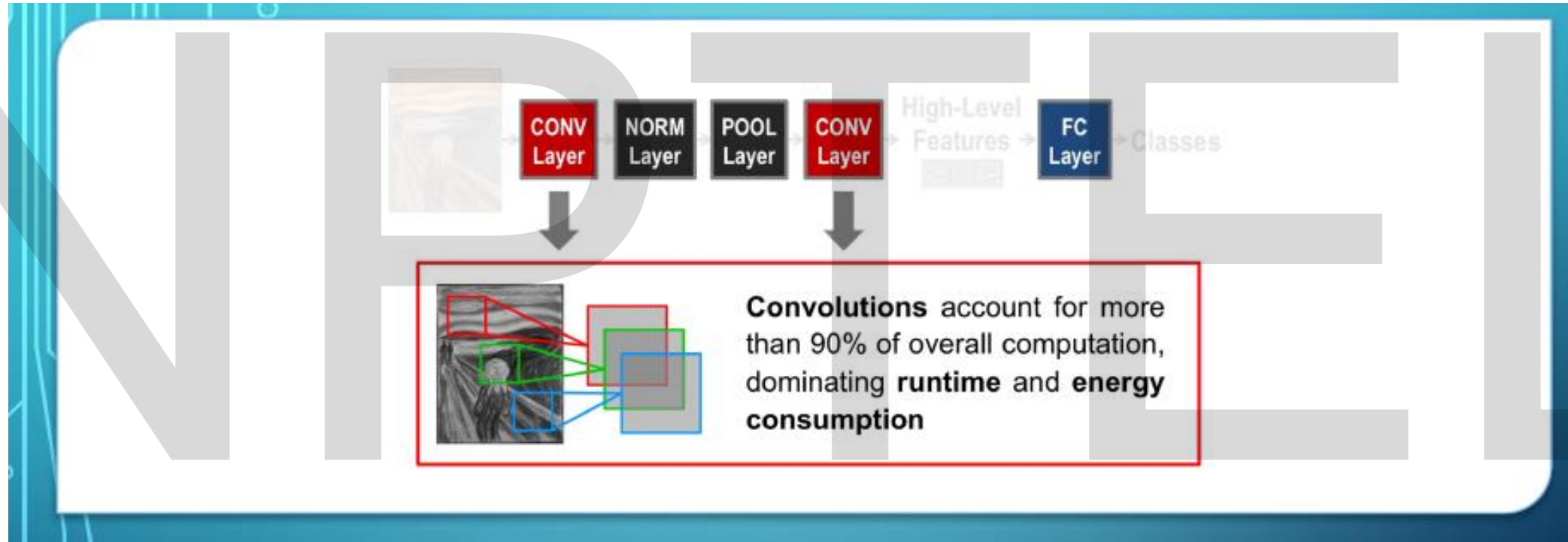


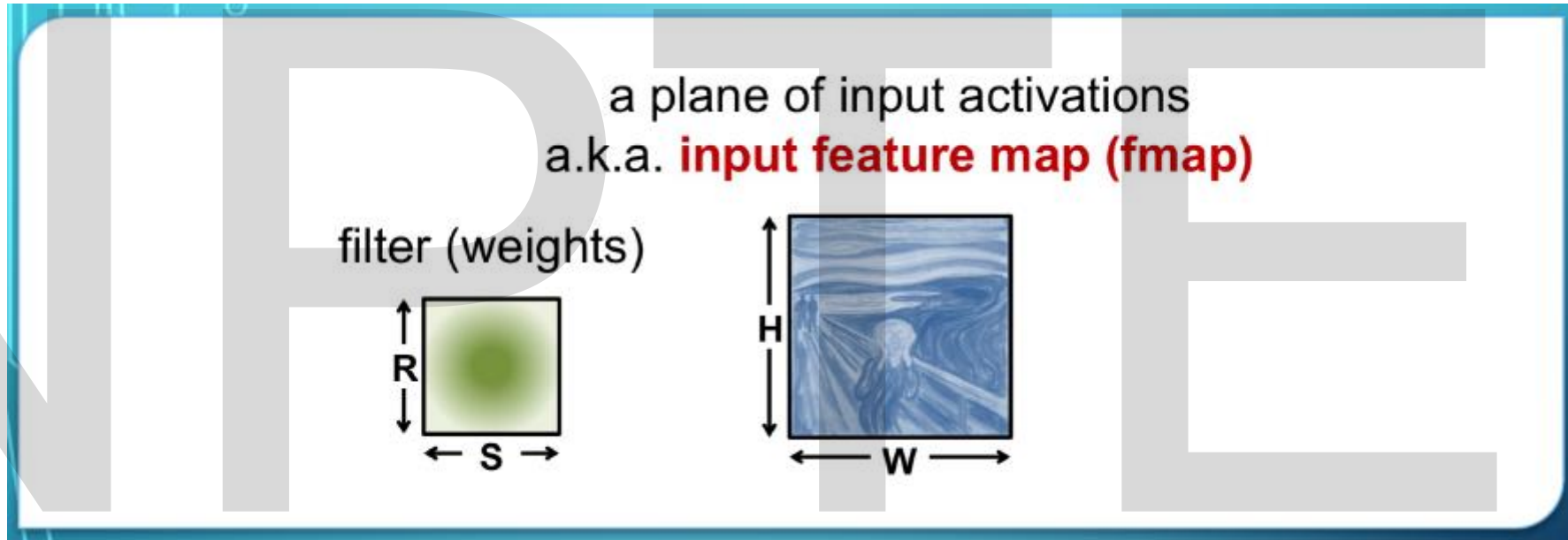
Deep CNNs



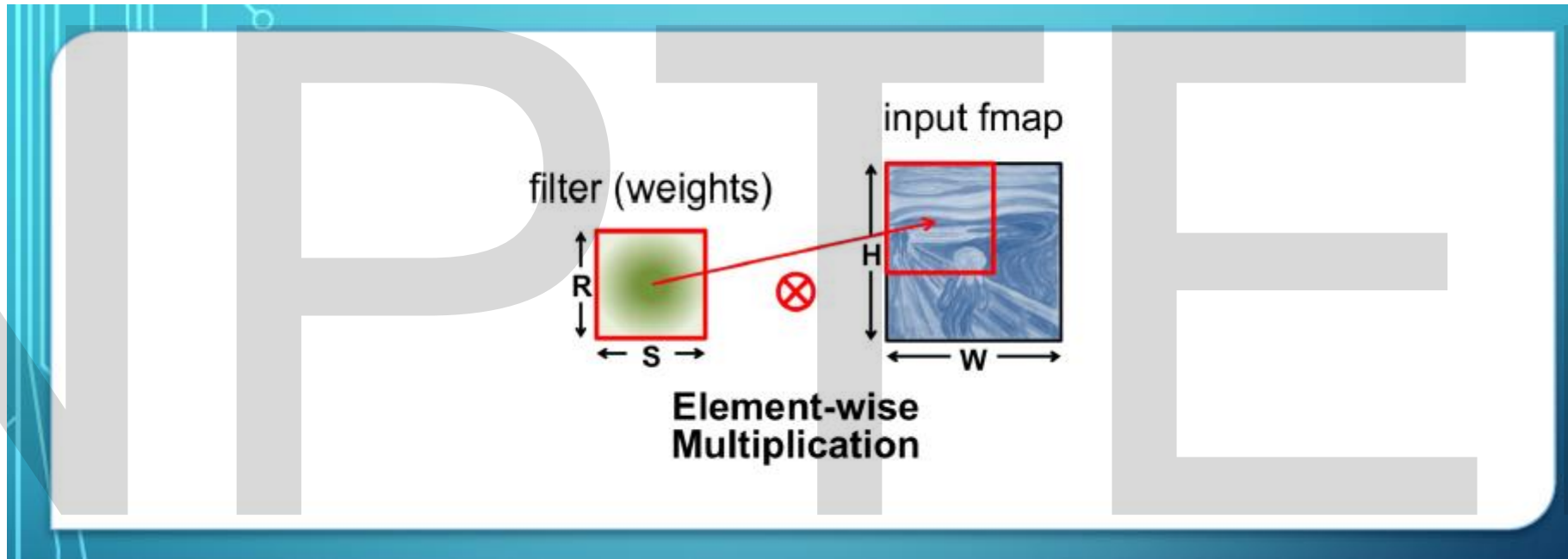
Deep CNNs



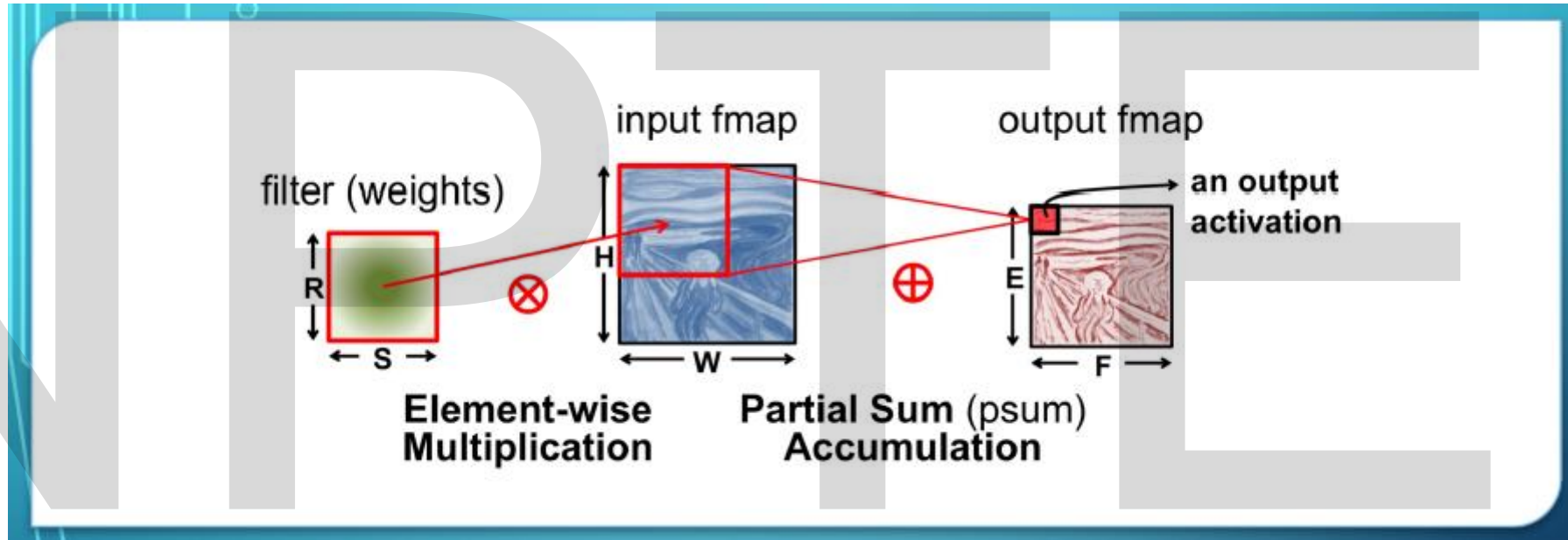
Convolution Process



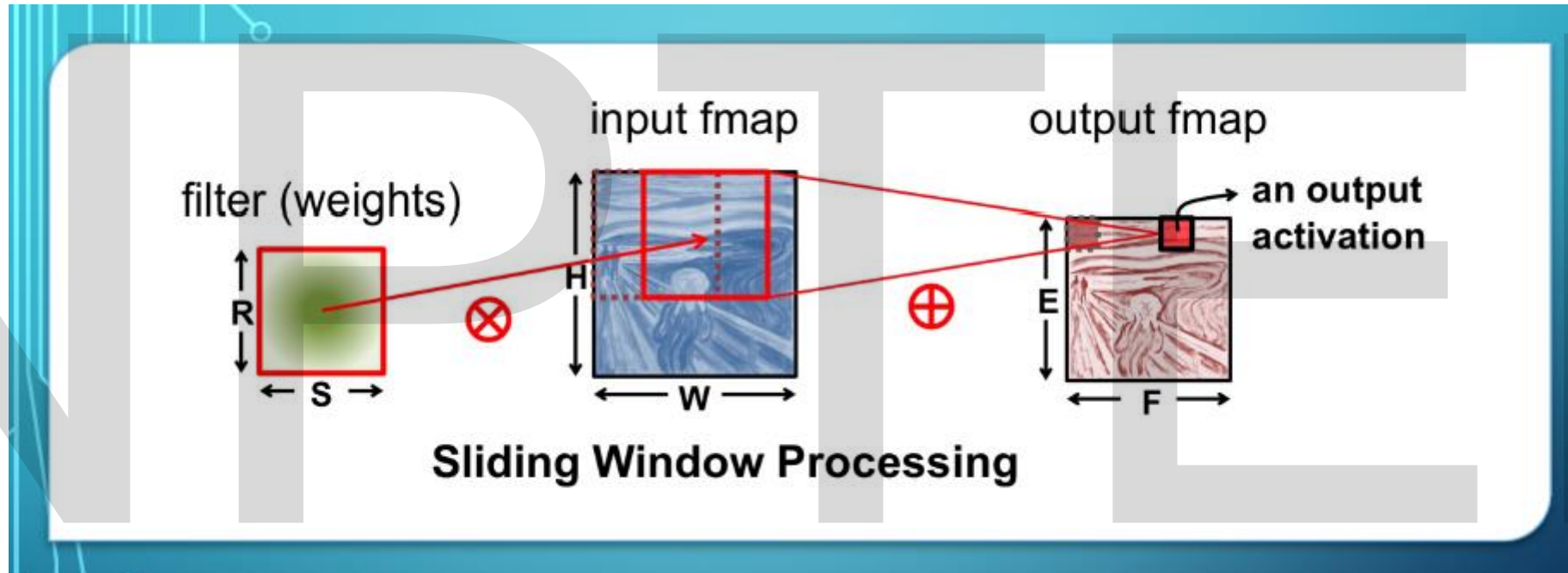
Convolution Process (Conv Layer)



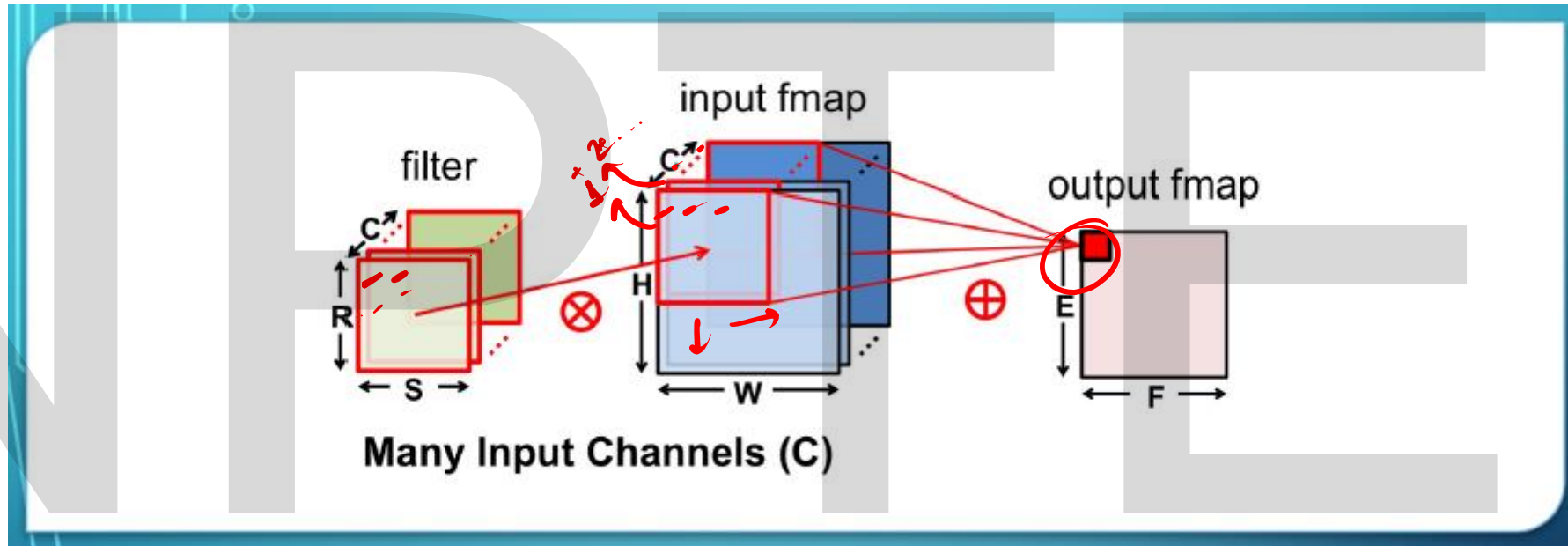
Conv Layer

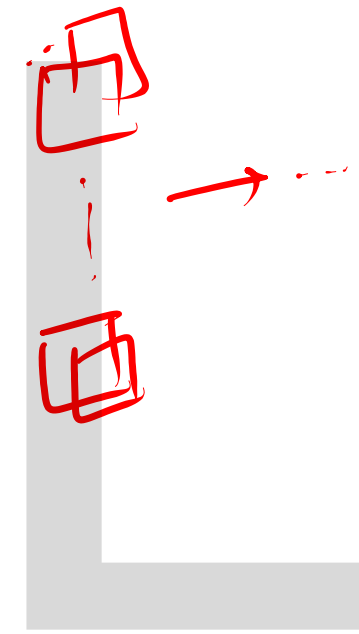


Conv Layer

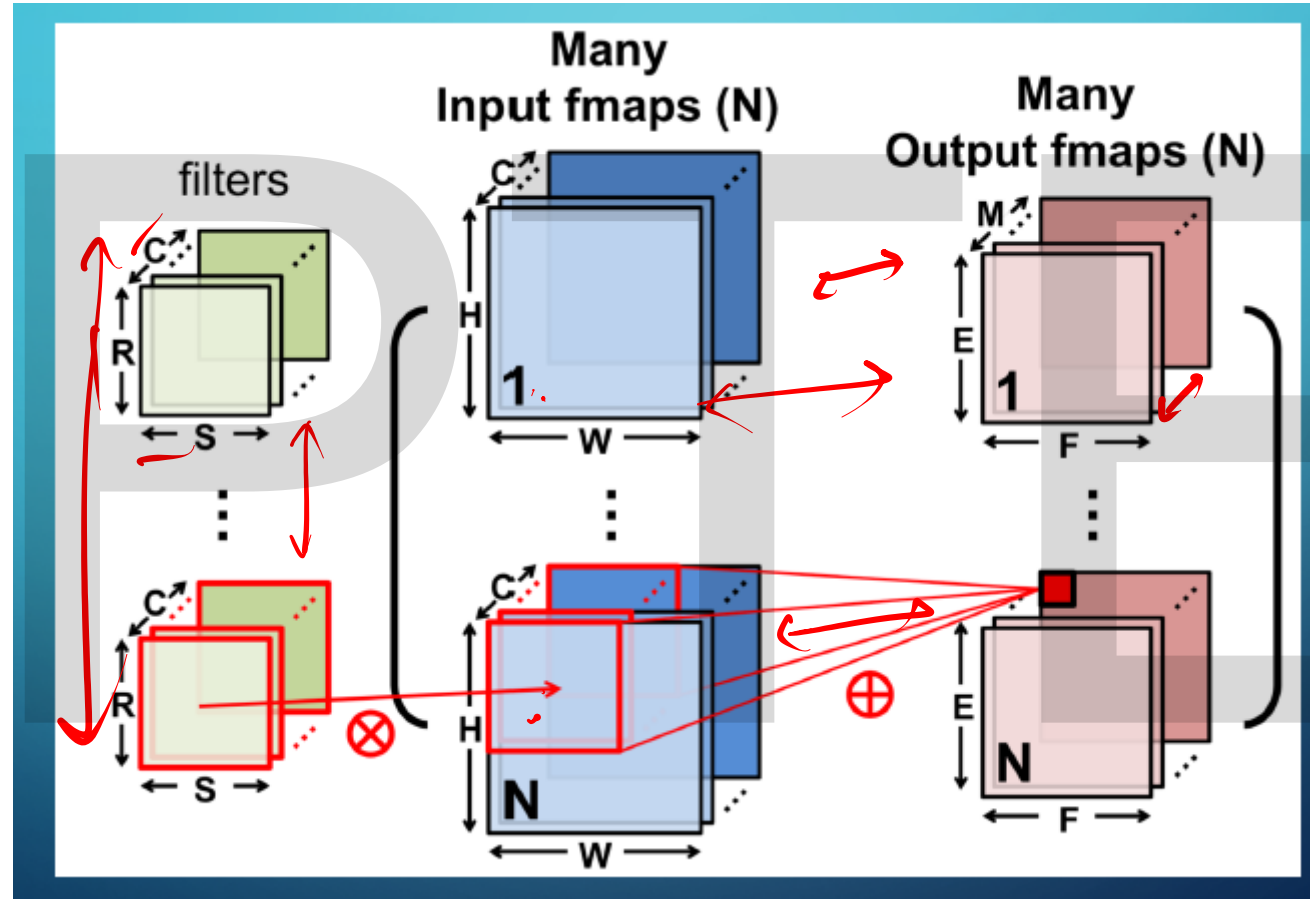


Conv Layer (Learning Across the Channel)





Conv Layer (Batch Multiple Inputs)

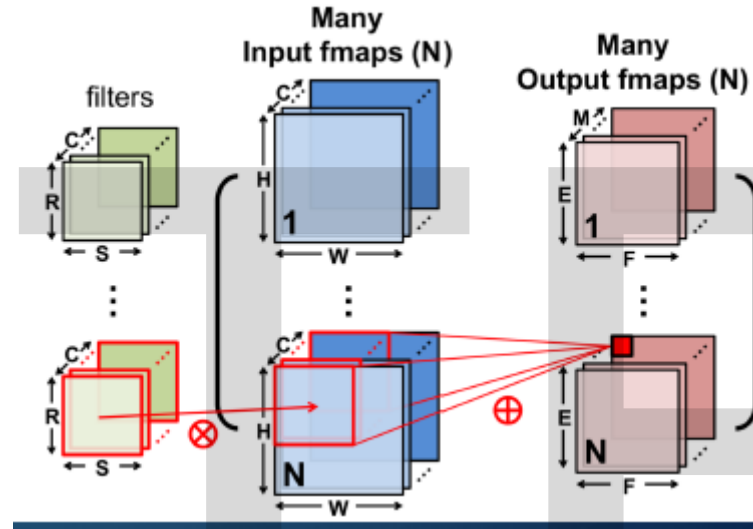


CNN Parameters

- N – Number of input fmaps/output fmaps (batch size)
- C – Number of 2-D input fmaps /filters (channels)
- H – Height of input fmap (activations)
- W – Width of input fmap (activations)
- R – Height of 2-D filter (weights)
- S – Width of 2-D filter (weights)
- M – Number of 2-D output fmaps (channels)
- E – Height of output fmap (activations)
- F – Width of output fmap (activations)

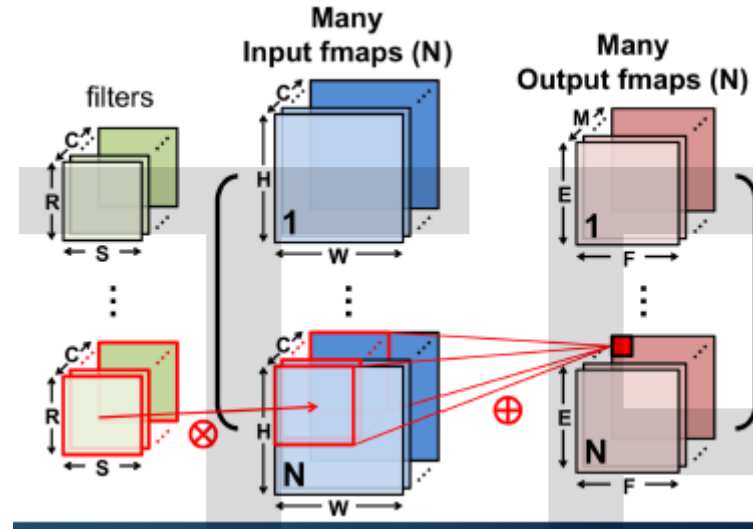
CNN Parameters

- **N** – Number of **input fmaps/output fmaps** (batch size)
- **C** – Number of 2-D **input fmaps /filters** (channels)
- **H** – Height of **input fmap** (activations)
- **W** – Width of **input fmap** (activations)
- **R** – Height of 2-D **filter** (weights)
- **S** – Width of 2-D **filter** (weights)
- **M** – Number of 2-D **output fmaps** (channels)
- **E** – Height of **output fmap** (activations)
- **F** – Width of **output fmap** (activations)



CNN Parameters

- **N** – Number of **input fmaps/output fmaps** (batch size)
- **C** – Number of 2-D **input fmaps /filters** (channels)
- **H** – Height of **input fmap** (activations)
- **W** – Width of **input fmap** (activations)
- **R** – Height of 2-D **filter** (weights)
- **S** – Width of 2-D **filter** (weights)
- **M** – Number of 2-D **output fmaps** (channels)
- **E** – Height of **output fmap** (activations)
- **F** – Width of **output fmap** (activations)



Output fmaps (O)

Input fmaps (I)

Biases (B)

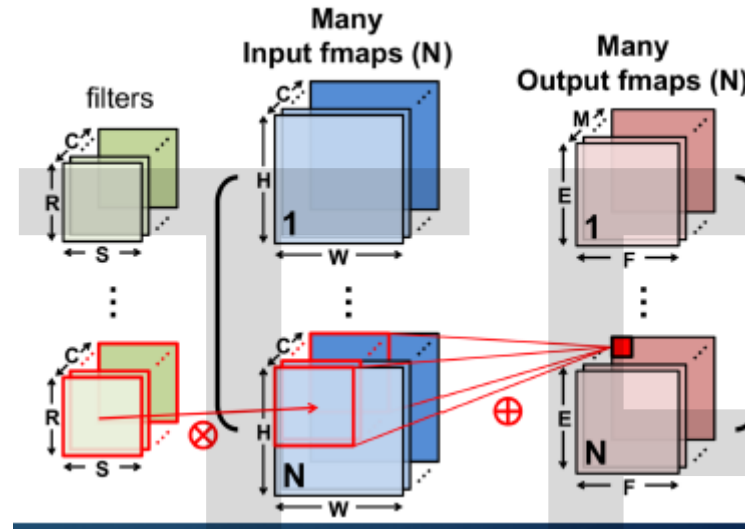
Filter weights (W)

$$O[n][m][x][y] = \text{Activation}(\underline{B[m]} + \sum_{i=0}^{R-1} \sum_{j=0}^{S-1} \sum_{k=0}^{C-1} \underline{I[n][k][Ux+i][Uy+j]} \times \underline{W[m][k][i][j]}),$$

$$0 \leq n < N, 0 \leq m < M, 0 \leq y < E, 0 \leq x < F,$$

CNN Parameters

- **N** – Number of **input fmaps/output fmaps** (batch size)
- **C** – Number of 2-D **input fmaps /filters** (channels)
- **H** – Height of **input fmap** (activations)
- **W** – Width of **input fmap** (activations)
- **R** – Height of 2-D **filter** (weights)
- **S** – Width of 2-D **filter** (weights)
- **M** – Number of 2-D **output fmaps** (channels)
- **E** – Height of **output fmap** (activations)
- **F** – Width of **output fmap** (activations)



Output fmaps (O)

Input fmaps (I)

Biases (B)

Filter weights (W)

$$O[n][m][x][y] = \text{Activation}(\underline{B[m]} + \sum_{i=0}^{R-1} \sum_{j=0}^{S-1} \sum_{k=0}^{C-1} \underline{I[n][k][Ux+i][Uy+j]} \times \underline{W[m][k][i][j]}),$$

$$0 \leq n < N, 0 \leq m < M, 0 \leq y < E, 0 \leq x < F,$$

Shape Parameter	Description
N	fmap batch size
M	# of filters / # of output fmap channels
C	# of input fmap/filter channels
H/W	input fmap height/width
R/S	filter height/width
E/F	output fmap height/width
U	convolution stride

Are you still with me?

